

Project Overview

Project overview:

Our group aims to demonstrate a low profile, flat form factor wheeled robot to augment the flow of materials around our chosen site, Ramatex's textile factory.

Currently, cage trolleys of materials are moved around by dedicated workers one at a time, with a low level of digital inventory management.

A few unique constraints present themselves in this case. Firstly, space on the factory floor is limited and in some places & walkways cramped. Our group determines that this limits the form factor of this system to under riding robots

Secondly, the factory utilises a fleet of non-standardised cage trolleys of slightly varying form factors in various dimensions.

Lastly the factory does not wish to spend large amounts of resources and man-hours retrofitting either the trolleys (Total est 2000 trolleys) or on structural retrofitting (Magnetic lines for line following, QR code for AGV).

Our solution aims to address these constraints and demonstrate it during the 30.007 project showcase.

A functionally downscaled (from est 180kg) aim of around 70kg load moved for the duration of a 1h demonstration was deemed achievable and chosen as a project aim.

Despite functional downscaling, the prototype still presents non-trivial power requirements. While feasible, our team wishes to check the soundness of our technical design process and prevent potential loss of resources (or limb) from electrical incidents.

Considering this factor, our team has decided we would like to discuss the electrical power details with Prof Teo.

Power distribution plan

Power distribution plan

Core components to be powered:

For movement:

- 120W, 24V, 5A nominal BLDC motor *4

For retractable arms:

- 60W, 12V, 5A nominal BLDC (stall current 10A) *2
- 6V 5A stall current servo motor *4

For control system:

- 5V, 5A Raspberry Pi 5 *1
- 3.3V 100mA ESP 32 *4 (est)
- Peripheral sensors (TBC)

Overall architecture

Key idea: Main +24V bus feeds separate branch families (drive motors with the regen clamp, secondary 12V motors, servo buck, the isolated Pi rail, and the MCU rail), and a common ground return structure.

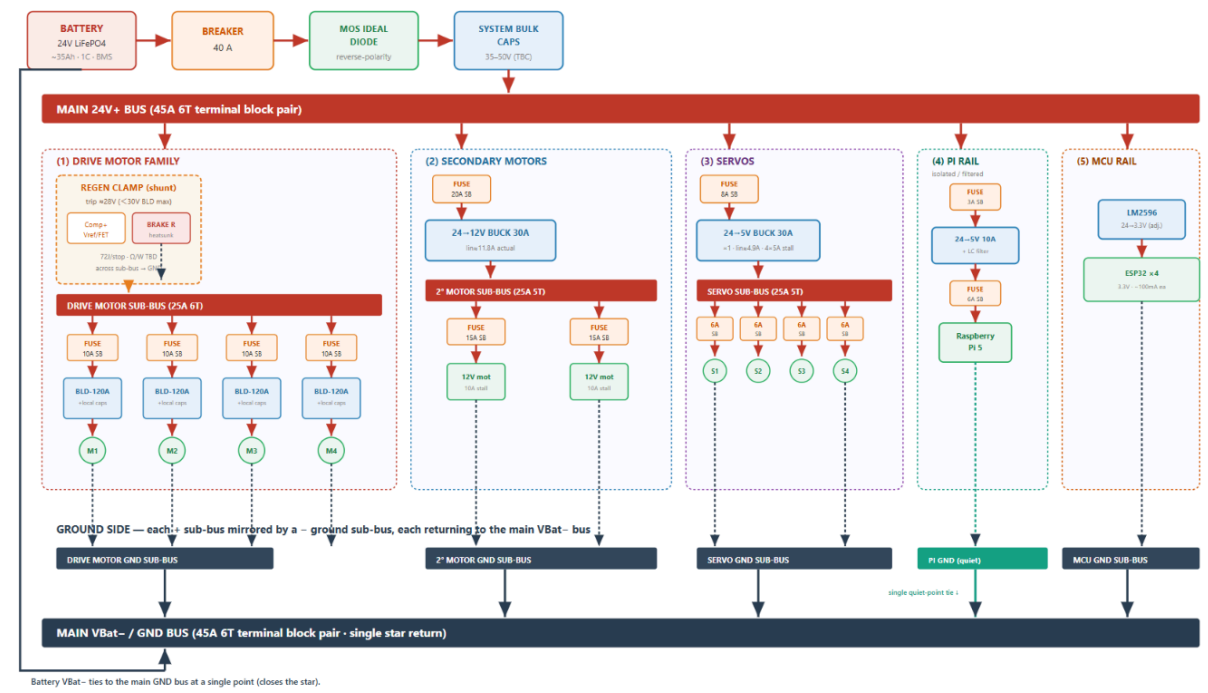


Fig 1: Overall system architecture (Generated by Claude)

Power distribution system proposed BOM:

24-12V 30A DC-DC step down buck *1 [for 2 60W BLDC]

24-5V 30A DC-DC step down buck *1 [for 4 servo motors]
24-5V 10A DC-DC step down buck *1 [for Pi]
LM2596 DC-DC step down buck *1 [for MCUs]
Regen clamp components (TBD)
45A 6 Terminal (T) Terminal block (TB) *2 [Main Bus]
45A 6T Jumper *2
25A 6T TB *2 [Drive motor sub Bus]
25A 6T Jumper *2
25A 5T TB *4 [2 for Secondary system motor sub Bus] [2 for Servo sub bus]
25A 5T Jumper *4
Logic component power sub bus (TBD)
18AWG cables *(tbc)
12AWG cable *(tbc)
8AWG cable *(tbc)
40A Circuit breaker *1
MOS Ideal Diode Module *1
20A slow blow fuse *1 [at 24-12V 30A buck input]
8A slow blow fuse *1 [at 24-5V 30A buck input]
3A slow blow fuse *1 [at 24-5V 10A buck input]
10A, slow blow fuses *10 (inc spares) [drive motor branches]
15A, slow blow fuses *10 (inc spares) [secondary motor branches]
6A, slow blow fuses *10 (inc spares) [for servo branches]
6A slow blow fuse *5 (inc spares) [for R pi]
Bulk capacitors (TBC) [system and motors]
Decoupling capacitors (TBC) [logic components power rail]

(A detailed BOM is included on the next Tab)

Costs

It's bad....

Testing plan

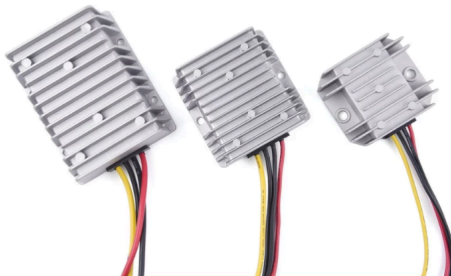
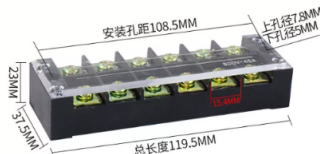
Test (Consult prof)

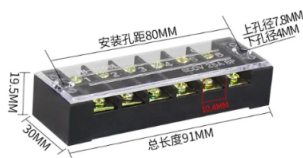
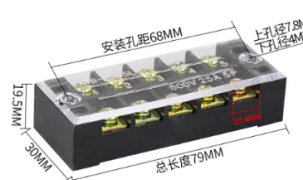
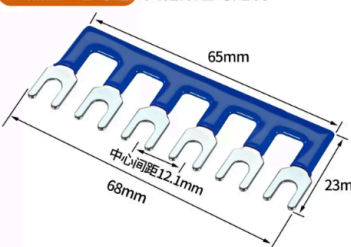
Queries

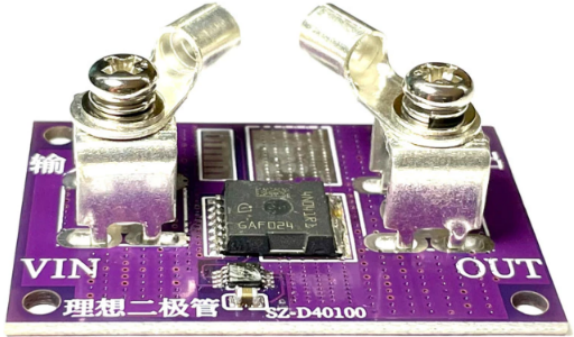
- Wires from dyson lab (was for power E). Can use?

Detailed BOM

BOM

Item	Qty	Purpose	Est Cost	Link	Notes																						
24-12V 30A DC-DC step down buck 24/36/48/60/72V to 12/13.8/24V DC Converter  3A/5A/10A/15A/20A/25A/30A/40A	1	for 2 60W BLDC	21.88	-	Tobsun																						
24-5V 30A DC-DC step down buck	1	for 4 servo motors		-																							
24-5V 10A DC-DC step down buck <table><tr><td>Image</td><td></td></tr><tr><td>Type</td><td>DC Buck Power Converter</td></tr><tr><td>Input Voltage Range</td><td>DC 12V / 24V</td></tr><tr><td>Output Voltage</td><td>DC 5V</td></tr><tr><td>Output Max. Current</td><td>3A 5A 10A 15A 20A 30A 40A</td></tr><tr><td>Output Max. Power</td><td>15W 25W 50W 75W 100W 150W 200W</td></tr><tr><td>Conversion Efficiency</td><td>≥94%</td></tr><tr><td>Waterproof Level</td><td>IP68</td></tr><tr><td>Working Temperature</td><td>-30℃ to +80℃</td></tr><tr><td>Size</td><td>67.5*58.4*23.5mm (2.66*2.3*0.93in) 74.4*74*31.1mm (2.93*2.91*1.22in) 100*80.2*38.4mm (3.94*3.16*1.51in)</td></tr><tr><td>Cable Length</td><td>About 15cm/0.49ft</td></tr></table>	Image		Type	DC Buck Power Converter	Input Voltage Range	DC 12V / 24V	Output Voltage	DC 5V	Output Max. Current	3A 5A 10A 15A 20A 30A 40A	Output Max. Power	15W 25W 50W 75W 100W 150W 200W	Conversion Efficiency	≥94%	Waterproof Level	IP68	Working Temperature	-30℃ to +80℃	Size	67.5*58.4*23.5mm (2.66*2.3*0.93in) 74.4*74*31.1mm (2.93*2.91*1.22in) 100*80.2*38.4mm (3.94*3.16*1.51in)	Cable Length	About 15cm/0.49ft	1	for Pi	12.57	-	Tobsun
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Cable Length	About 15cm/0.49ft																										
LM2596 DC-DC step down buck	1	for MCUs	0	-	In inventory																						
45A 6 Terminal (T) Terminal block (TB) TB4506 	2	Main Bus	12.25	-	Inc shipping																						

25A 6T TB TB2506 	2	Drive motor sub Bus			
25A 5T TB TB2505 	4	2 for Secondary system motor sub Bus 2 for Servo sub bus			
45A 6T Jumper TB-4506-6位 红黑蓝 3色可选 其他颜色可定制 	2	Main Bus		=	
25A 6T Jumper TB-2506-6位 红黑蓝 3色可选 其他颜色可定制 	2	Drive motor sub Bus			

25A 5T Jumper  红黑蓝 3色可选 其他颜色可定制 53mm 23mm 中心间距12.1mm 56mm	4	2 for Secondary system motor sub Bus 2 for Servo sub bus			
Logic component power sub bus	TBD	Logic component sub bus		-	SPECIFICS TBD
18AWG cables	-	For internal wiring	0	-	From Dyson Lab
12AWG cable	-	For internal wiring	0	-	
8AWG cable	-	For battery to main bus	0	-	
40A Circuit breaker	1	Battery to Main bus connection	8		
MOS Ideal Diode Module (SZ-D40100)  VIN OUT 理想二极管 SZ-D40100	1	Battery to Main bus connection	8.99	-	
20A slow blow fuse	5	at 24-12V 30A buck input		-	Including spares
8A slow blow fuse	5	at 24-5V 30A buck input		-	
3A slow blow fuse	5	at 24-5V 10A buck input		-	
10A slow blow fuses	10	drive motor branches		-	
15A slow blow fuses	10	secondary motor branches		-	
6A slow blow fuses	10	for servo branches		-	

6A slow blow fuse	5	for R pi		-	
Bulk capacitors	TBC	system and motors		-	SPECIFICS TBD
Decoupling capacitors	TBC	logic components power rail		-	SPECIFICS TBD
Regen clamp components	TBD	Clamp with break resistor		-	SPECIFICS TBD

Tab 4

New design

Main 36V bus:

[36/48 - 12V 25A](#) 18.33 (Taobao)

[36/48 - 24V 30A](#) 35.69 (Taobao)

[36/48 - 24V 20A](#) 17.36 (Taobao)

TB4504 *2 (Taobao)

TB4506 *2 (Taobao)

TB4505 *2 (Taobao)

TB2506 *2 (Taobao)

Mos Reverse current protection unchanged 8.99 (Shopee)

Fuses and holder (Taobao)

Standard blade fuse rated to 33V, can use for sub branched but not main bus branch/ battery break

HW4

My contributions

I have been working and iterating on a power system to power all the components of the prototype. With the help of friends who are more experienced in the field, professors, and Claude AI, I have settled on a system that I think is good enough for a first prototype. There have been 3 iterations of the idea of the system architecture so far:

Iteration 1 (Fig 1) was a pathfinder as I learned about the different components we need for the power system to work.

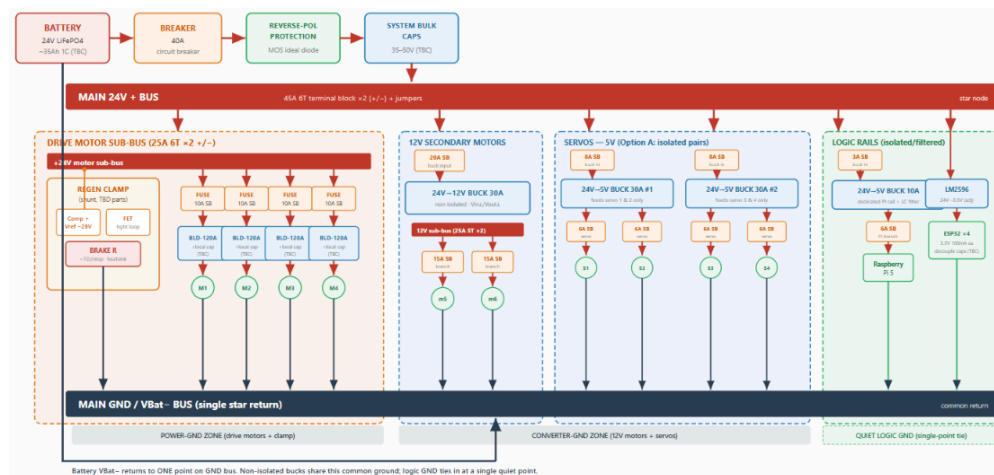


Figure 1: Iteration 1 of Overall PDS system architecture and core components

Iteration 2 (Fig 2) refined the ideas of iteration 1 and added in consideration for more FOS on the power rating of components like Terminal blocks, Buck converters, etc. Future consideration for cable management was also made by compartmentalising each of the different branches.

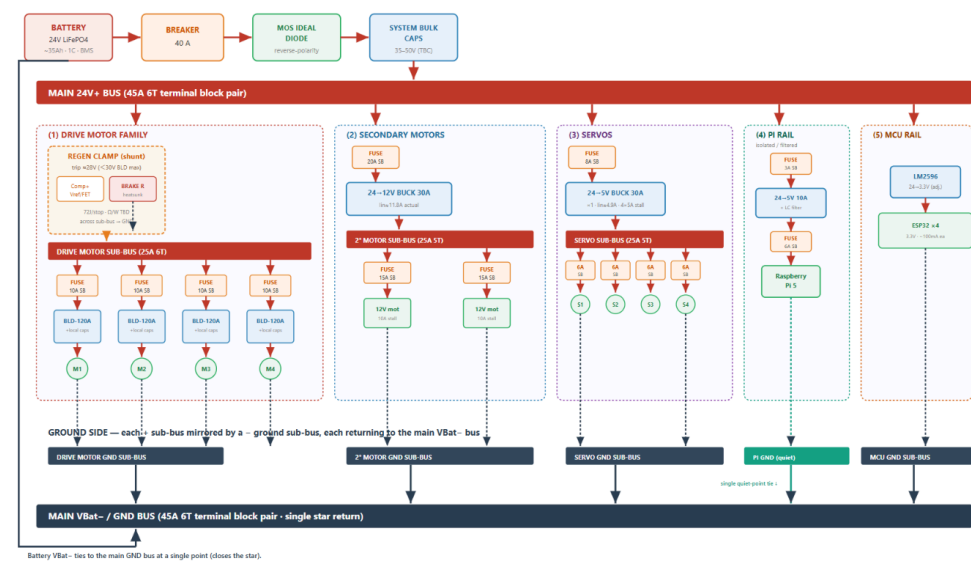
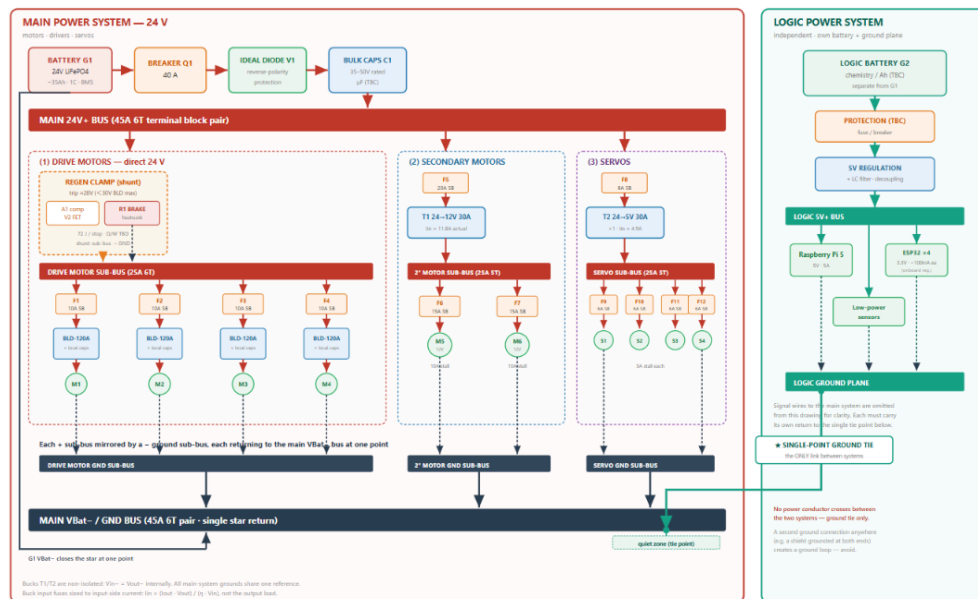


Figure 2: Iteration 2 of Overall PDS system architecture and core components

Iteration 3 (Fig 3) makes several more drastic changes from iteration 2. Following a consult with Prof Teo Tee Hui for his domain knowledge in designing power systems and power electronics, the changes made were:

- Running a 36V instead of 24V battery for more constant voltage supply
- Isolating the power system of the Logic and control units from the main power distribution system
- Addition of a E stop and safety relay to the main Power system (not shown in Fig 3) for greater safety in case of emergency.

*Figure 1: Iteration 3 of Overall PDS system architecture and core components*

Based on Iteration 3, a BOM was generated and the parts procured from various sources. While basically complete, the system is still missing the bulk capacitor banks necessary to prevent unwanted voltage drops on the power rails during move off and other events of higher power draw.

The next step after the parts arrive will be to test each separately to verify performance truthful to specifications. Following which steady integration and testing of more and more parts can be carried out.

Revised concept requirements

	<u>Concept requirement</u>	<u>Why is it important</u>
1	Cable management and cable harness design	With the number of connections and components that are present in our circuit, a lot of wiring and cables are expected. Messy cables are both hard to work with and a hazard to the safe operation of the prototype.
2	Thermal monitoring and control	While considered as “low power” on an industrial scale, there is still high potential for heat buildup to be an issue with all the components we are intending to fit into a low profile machine. Monitoring (sensors) and control (cooling) are ways we can help to prevent incidence and loss of equipment and more importantly personnel
3	PCB design	Some of the components can be combined into PCBs to make the system more robust and neat.
4	On site fire safety	See point 2